Aurora: Module Relay x8

Stakeholders Answers

Prepared for

Technical Staff

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Disclosure is prohibited

Revision History

Version	Date	Author	Description
0.1.0	2026-07-02	Martin Pellizzer	Starting version

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Answers

Answer 1: System Architect

Question

What exactly is the responsibility of the Relay Module within Aurora?

Answer

The Relay Module shall reliably and deterministically execute relay-output commands received from the CORE, provide the CORE with output status and module diagnostics, and transition its outputs to defined safe states when required by system-level fault conditions.

Field	Value
Stakeholder ID	1
Question ID	1
Answer ID	1
Stakeholder Role	System Architect
Question Text	What exactly is the responsibility of the Relay Module within Aurora?
Answer Text	The Relay Module shall reliably and deterministically execute relay-output commands received from the CORE, provide the CORE with output status and module diagnostics, and transition its outputs to defined safe states when required by system-level fault conditions.

Answer 2: System Architect

Question

Which functions of the overall Aurora system are allocated specifically to the Relay Module?

Answer

The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.

Field	Value
Stakeholder ID	1

Field	Value
Question ID	2
Answer ID	2
Stakeholder Role	System Architect
Question Text	Which functions of the overall Aurora system are allocated specifically to the Relay Module?
Answer Text	The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.

Answer 3: System Architect

Question

What behavior does the system require from each of the eight relay outputs?

Answer

Each output shall independently assume the ON or OFF state commanded by the CORE within the specified response time, while preventing an individual channel fault from affecting the other channels and maintaining the output's defined safe behavior under fault conditions.

Field	Value
Stakeholder ID	1
Question ID	3
Answer ID	3
Stakeholder Role	System Architect
Question Text	What behavior does the system require from each of the eight relay outputs?
Answer Text	Each output shall independently assume the ON or OFF state commanded by the CORE within the specified response time, while preventing an individual channel fault from affecting the other channels and maintaining the output's defined safe behavior under fault conditions.

Answer 4: System Architect

Question

What must happen to every relay output when communication with the CORE is lost?

Answer

Upon detecting a communication timeout with the CORE, every relay output shall transition to its predefined safe state, which shall be OFF unless a system-level hazard analysis explicitly requires another state.

Field	Value
Stakeholder ID	1
Question ID	4
Answer ID	4
Stakeholder Role	System Architect
Question Text	What must happen to every relay output when communication with the CORE is lost?
Answer Text	Upon detecting a communication timeout with the CORE, every relay output shall transition to its predefined safe state, which shall be OFF unless a system-level hazard analysis explicitly requires another state.

Answer 5: System Architect

Question

What is the required system behavior if the Relay Module itself fails?

Answer

Any Relay Module failure that could compromise safe operation shall cause the affected outputs to transition to their defined safe states, prevent unintended activation, and provide sufficient fault information to the CORE to initiate the appropriate Aurora system fault response.

Field	Value
Stakeholder ID	1
Question ID	5
Answer ID	5
Stakeholder Role	System Architect
Question Text	What is the required system behavior if the Relay Module itself fails?

Field	Value
Answer Text	Any Relay Module failure that could compromise safe operation shall cause the affected outputs to transition to their defined safe states, prevent unintended activation, and provide sufficient fault information to the CORE to initiate the appropriate Aurora system fault response.

Answer 6: CORE Firmware Engineer

Question

Which responsibilities belong to the CORE and which responsibilities belong to the Relay Module?

Answer

The CORE shall own all system-level decisions, sequencing, interlocks, operating modes and desired relay states, while the Relay Module shall execute those commands deterministically, manage its eight physical driver outputs, perform local hardware diagnostics, and report its status and faults to the CORE.

Field	Value
Stakeholder ID	2
Question ID	6
Answer ID	6
Stakeholder Role	CORE Firmware Engineer
Question Text	Which responsibilities belong to the CORE and which responsibilities belong to the Relay Module?
Answer Text	The CORE shall own all system-level decisions, sequencing, interlocks, operating modes and desired relay states, while the Relay Module shall execute those commands deterministically, manage its eight physical driver outputs, perform local hardware diagnostics, and report its status and faults to the CORE.

Answer 7: CORE Firmware Engineer

Question

What commands must the CORE be able to send to the Relay Module?

Answer

The CORE shall be able to command each of the eight outputs independently, initialize and reset the module, request status and diagnostics, and synchronize or restore the required output states after startup or communication recovery.

Field	Value
Stakeholder ID	2
Question ID	7
Answer ID	7
Stakeholder Role	CORE Firmware Engineer
Question Text	What commands must the CORE be able to send to the Relay Module?
Answer Text	The CORE shall be able to command each of the eight outputs independently, initialize and reset the module, request status and diagnostics, and synchronize or restore the required output states after startup or communication recovery.

Answer 8: CORE Firmware Engineer

Question

What information must the Relay Module provide back to the CORE?

Answer

The Relay Module shall report the state of all eight outputs, module health, individual channel faults, communication status, relevant electrical/thermal diagnostics, firmware/hardware identification, and any condition preventing reliable output control.

Field	Value
Stakeholder ID	2
Question ID	8
Answer ID	8
Stakeholder Role	CORE Firmware Engineer
Question Text	What information must the Relay Module provide back to the CORE?
Answer Text	The Relay Module shall report the state of all eight outputs, module health, individual channel faults, communication status, relevant electrical/thermal diagnostics, firmware/hardware identification, and any condition preventing reliable output control.

Answer 9: CORE Firmware Engineer

Question

How should the CORE detect that communication with the Relay Module has been lost?

Answer

The CORE shall use a supervised communication mechanism with a defined response timeout and consecutive failed transactions, declaring the Relay Module unavailable when the configured communication-loss threshold is exceeded.

Field	Value
Stakeholder ID	2
Question ID	9
Answer ID	9
Stakeholder Role	CORE Firmware Engineer
Question Text	How should the CORE detect that communication with the Relay Module has been lost?
Answer Text	The CORE shall use a supervised communication mechanism with a defined response timeout and consecutive failed transactions, declaring the Relay Module unavailable when the configured communication-loss threshold is exceeded.

Answer 10: CORE Firmware Engineer

Question

What should the CORE do when communication with the Relay Module is lost?

Answer

The CORE shall immediately declare a Relay Module communication fault, prevent further normal relay commands, transition Aurora to the appropriate safe system state, notify the HMI, and automatically re-synchronize the required relay states only after reliable communication has been restored.

Field	Value
Stakeholder ID	2
Question ID	10

Field	Value
Answer ID	10
Stakeholder Role	CORE Firmware Engineer
Question Text	What should the CORE do when communication with the Relay Module is lost?
Answer Text	The CORE shall immediately declare a Relay Module communication fault, prevent further normal relay commands, transition Aurora to the appropriate safe system state, notify the HMI, and automatically re-synchronize the required relay states only after reliable communication has been restored.

Answer 11: Electrical Engineer

Question

What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?

Answer

Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.

Field	Value
Stakeholder ID	3
Question ID	11
Answer ID	11
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?
Answer Text	Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.

Answer 12: Electrical Engineer

Question

What are the electrical characteristics of the physical relay coils connected to each of the eight driver outputs?

Answer

Each relay coil shall have a defined nominal voltage, allowable voltage range, nominal and maximum current, resistance/inductance, inrush behavior, continuous-duty rating, and release/operate characteristics that are fully compatible with its corresponding driver.

Field	Value
Stakeholder ID	3
Question ID	12
Answer ID	12
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical characteristics of the physical relay coils connected to each of the eight driver outputs?
Answer Text	Each relay coil shall have a defined nominal voltage, allowable voltage range, nominal and maximum current, resistance/inductance, inrush behavior, continuous-duty rating, and release/operate characteristics that are fully compatible with its corresponding driver.

Answer 13: Electrical Engineer

Question

What is the required input supply voltage range, including tolerances and transient conditions?

Answer

The module shall operate correctly over the complete specified nominal supply range, including component tolerances, steady-state variation, startup conditions, brownouts, overvoltage, ripple, and transient disturbances expected in the Aurora electrical cabinet.

Field	Value
Stakeholder ID	3
Question ID	13
Answer ID	13

Field	Value
Stakeholder Role	Electrical Engineer
Question Text	What is the required input supply voltage range, including tolerances and transient conditions?
Answer Text	The module shall operate correctly over the complete specified nominal supply range, including component tolerances, steady-state variation, startup conditions, brownouts, overvoltage, ripple, and transient disturbances expected in the Aurora electrical cabinet.

Answer 14: Electrical Engineer

Question

What protection is required to handle the inductive voltage transient generated when a relay coil is de-energized?

Answer

Each driver output shall incorporate appropriately sized suppression, such as a flyback diode or TVS/clamp network, to limit the coil-generated transient to a safe level without exceeding driver ratings or causing unacceptable relay release-time delays.

Field	Value
Stakeholder ID	3
Question ID	14
Answer ID	14
Stakeholder Role	Electrical Engineer
Question Text	What protection is required to handle the inductive voltage transient generated when a relay coil is de-energized?
Answer Text	Each driver output shall incorporate appropriately sized suppression, such as a flyback diode or TVS/clamp network, to limit the coil-generated transient to a safe level without exceeding driver ratings or causing unacceptable relay release-time delays.

Answer 15: Electrical Engineer

Question

What electrical behavior is required during power loss, brownout, and restoration of power?

Answer

During power loss or an out-of-range brownout, all relay outputs shall transition to and remain in their defined safe state without unintended activation, and after power restoration the module shall initialize deterministically with outputs disabled until valid control from the CORE is established.

Field	Value
Stakeholder ID	3
Question ID	15
Answer ID	15
Stakeholder Role	Electrical Engineer
Question Text	What electrical behavior is required during power loss, brownout, and restoration of power?
Answer Text	During power loss or an out-of-range brownout, all relay outputs shall transition to and remain in their defined safe state without unintended activation, and after power restoration the module shall initialize deterministically with outputs disabled until valid control from the CORE is established.

Answer 16: Safety Engineer

Question

What safety functions, if any, are assigned specifically to the Relay Module?

Answer

The Relay Module shall locally enforce the safety-related output constraints and fail-safe behavior defined for its eight relay channels, detect relevant hardware and output faults, and transition affected outputs to their predefined safe states independently of higher-level system decisions.

Field	Value
Stakeholder ID	4
Question ID	16
Answer ID	16
Stakeholder Role	Safety Engineer
Question Text	What safety functions, if any, are assigned specifically to the Relay Module?

Field	Value
Answer Text	The Relay Module shall locally enforce the safety-related output constraints and fail-safe behavior defined for its eight relay channels, detect relevant hardware and output faults, and transition affected outputs to their predefined safe states independently of higher-level system decisions.

Answer 17: Safety Engineer

Question

What is the required safe state for each of the eight relay outputs?

Answer

Each of the eight relay outputs shall have a formally defined safe state based on its associated hazard analysis, with OFF/de-energized as the default safe state unless a documented safety analysis requires a specific output to remain energized.

Field	Value
Stakeholder ID	4
Question ID	17
Answer ID	17
Stakeholder Role	Safety Engineer
Question Text	What is the required safe state for each of the eight relay outputs?
Answer Text	Each of the eight relay outputs shall have a formally defined safe state based on its associated hazard analysis, with OFF/de-energized as the default safe state unless a documented safety analysis requires a specific output to remain energized.

Answer 18: Safety Engineer

Question

What shall happen to each relay output when the Relay Module detects an internal fault?

Answer

When the Relay Module detects an internal fault capable of compromising safe operation, it shall immediately prevent unsafe output activation, force all affected outputs to their defined safe states, latch or report the fault as required, and prevent reactivation until the fault-clear and reauthorization conditions are satisfied.

Field	Value
Stakeholder ID	4
Question ID	18
Answer ID	18
Stakeholder Role	Safety Engineer
Question Text	What shall happen to each relay output when the Relay Module detects an internal fault?
Answer Text	When the Relay Module detects an internal fault capable of compromising safe operation, it shall immediately prevent unsafe output activation, force all affected outputs to their defined safe states, latch or report the fault as required, and prevent reactivation until the fault-clear and reauthorization conditions are satisfied.

Answer 19: Safety Engineer

Question

What shall happen to each relay output when communication with the CORE is lost?

Answer

When communication with the CORE is lost beyond the defined safety timeout, the Relay Module shall autonomously transition every output to its predefined safe state, maintain that state, report the communication fault when communication is restored, and require explicit reauthorization before returning to normal operation.

Field	Value
Stakeholder ID	4
Question ID	19
Answer ID	19
Stakeholder Role	Safety Engineer
Question Text	What shall happen to each relay output when communication with the CORE is lost?
Answer Text	When communication with the CORE is lost beyond the defined safety timeout, the Relay Module shall autonomously transition every output to its predefined safe state, maintain that state, report the communication fault when communication is restored, and require explicit reauthorization before returning to normal operation.

Answer 20: Safety Engineer

Question

Which safety decisions must be made by the CORE, and which must be enforced locally by the Relay Module?

Answer

The CORE shall make all system-level safety decisions, operating-mode and shutdown decisions, while the Relay Module shall locally enforce output-level safety constraints, watchdog/communication-loss responses, internal fault handling, and the defined safe-state behavior without relying solely on the CORE.

Field	Value
Stakeholder ID	4
Question ID	20
Answer ID	20
Stakeholder Role	Safety Engineer
Question Text	Which safety decisions must be made by the CORE, and which must be enforced locally by the Relay Module?
Answer Text	The CORE shall make all system-level safety decisions, operating-mode and shutdown decisions, while the Relay Module shall locally enforce output-level safety constraints, watchdog/communication-loss responses, internal fault handling, and the defined safe-state behavior without relying solely on the CORE.

Answer 21: Process Engineer

Question

What process functions are controlled or influenced by the eight relay outputs of the Relay Module?

Answer

The eight relay outputs shall control or influence the ozone-generation process by operating the required process equipment and auxiliaries—such as ozone-generation power, cooling, gas/air or oxygen supply, valves, purge/ventilation, and other process actuators—according to the defined process sequence and interlocks.

Field	Value
Stakeholder ID	5
Question ID	21
Answer ID	21
Stakeholder Role	Process Engineer
Question Text	What process functions are controlled or influenced by the eight relay outputs of the Relay Module?
Answer Text	The eight relay outputs shall control or influence the ozone-generation process by operating the required process equipment and auxiliaries—such as ozone-generation power, cooling, gas/air or oxygen supply, valves, purge/ventilation, and other process actuators—according to the defined process sequence and interlocks.

Answer 22: Process Engineer

Question

What is the required sequence of operations for starting ozone generation?

Answer

The startup sequence shall verify all required permissives, establish the required gas flow and supporting utilities, enable cooling and other auxiliaries, confirm their healthy operating feedback, energize the ozone-generation system, and only then transition Aurora to the normal ozone-production state.

Field	Value
Stakeholder ID	5
Question ID	22
Answer ID	22
Stakeholder Role	Process Engineer
Question Text	What is the required sequence of operations for starting ozone generation?
Answer Text	The startup sequence shall verify all required permissives, establish the required gas flow and supporting utilities, enable cooling and other auxiliaries, confirm their healthy operating feedback, energize the ozone-generation system, and only then transition Aurora to the normal ozone-production state.

Answer 23: Process Engineer

Question

What is the required sequence of operations for stopping ozone generation?

Answer

The shutdown sequence shall first disable ozone generation, maintain the required cooling and gas-flow/purge functions for the specified post-run period, safely de-energize the remaining auxiliaries in the defined order, and finally confirm that the process has reached the stopped state.

Field	Value
Stakeholder ID	5
Question ID	23
Answer ID	23
Stakeholder Role	Process Engineer
Question Text	What is the required sequence of operations for stopping ozone generation?
Answer Text	The shutdown sequence shall first disable ozone generation, maintain the required cooling and gas-flow/purge functions for the specified post-run period, safely de-energize the remaining auxiliaries in the defined order, and finally confirm that the process has reached the stopped state.

Answer 24: Process Engineer

Question

What conditions must be satisfied before ozone generation is permitted to start?

Answer

Ozone generation shall be permitted only when all required utilities are available, gas flow and cooling are within their specified limits, required valves and auxiliaries are in their correct states, no active safety or process interlock is violated, all required feedback signals are valid, and the system is in an authorized operating mode.

Field	Value
Stakeholder ID	5

Field	Value
Question ID	24
Answer ID	24
Stakeholder Role	Process Engineer
Question Text	What conditions must be satisfied before ozone generation is permitted to start?
Answer Text	Ozone generation shall be permitted only when all required utilities are available, gas flow and cooling are within their specified limits, required valves and auxiliaries are in their correct states, no active safety or process interlock is violated, all required feedback signals are valid, and the system is in an authorized operating mode.

Answer 25: Process Engineer

Question

What process conditions require ozone generation to stop immediately?

Answer

Ozone generation shall stop immediately whenever a condition capable of creating an unsafe or uncontrolled process state is detected, including loss of required gas flow or cooling, critical equipment failure, hazardous process-variable limits, loss of required safety permissives, emergency shutdown, or any other defined critical interlock.

Field	Value
Stakeholder ID	5
Question ID	25
Answer ID	25
Stakeholder Role	Process Engineer
Question Text	What process conditions require ozone generation to stop immediately?
Answer Text	Ozone generation shall stop immediately whenever a condition capable of creating an unsafe or uncontrolled process state is detected, including loss of required gas flow or cooling, critical equipment failure, hazardous process-variable limits, loss of required safety permissives, emergency shutdown, or any other defined critical interlock.